FRE 6103 Valuation – Midterm Study Guide

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The midterm is in week 7 (the syllabus lists Oct 16–18; check your section's day) and covers weeks 1–6. Prof. Shimko hands out a clean copy of the formula cheat sheet with the exam, so focus on knowing when to use each formula and how to set up the GVE, not on memorizing formulas. Bring a calculator with no internet connection.

Week 1 – Present value and the General Valuation Equation

PV is only one model of value. It is reliable only when the cash flows could be sold today at fair value. The GVE is the course's main tool, so be able to set it up for any cash flow pattern.

Material: Lecture 1A (Present Value Fails), Lecture 1B (MBE Brief), Valuation Principles Ch 1–2.

Discounting conventions

- n periods, discrete: V0 = Σ Ci / (1+r)^i
- Continuous compounding: V0 = Σ Ci e^(−ri). Continuous payments: V0 = ∫ Ct e^(−rt) dt
- Effective annual rate: 1 + EAR = (1 + r/k)^k. Bonds use simple division (r/k per period), not compounding.
- Corporate analysis uses discrete discounting. Derivatives use continuous discounting.

Why PV can "fail"

- A person can rationally prefer a lower-PV option because of cash constraints, high-interest debt, a better investment, or default risk. Example: $10,000 in 1 year (PV $9,524) vs $11,000 in 2 years (PV $9,977) at 5%.
- PV quietly assumes the cash flows are marketable. If they can be sold now at the market rate, take the higher PV.
- Other weaknesses: sensitivity to the discount rate, forecast error, terminal value, no optionality, and risk handled crudely through the discount rate.
- California insurance quiz: posting the $30,000 bond saves $1,500 a year in expected loss. But insurance is better if your opportunity cost of capital is above 10% (1,500 / 30,000 = 5% plus the 5% the bond pays).

The GVE ("most beautiful equation")

Required return = expected return. Time value plus risk compensation equals expected capital gain plus expected cash flow:

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rV_0 + k\sigma = \left(E[V_1] - V_0\right) + E[C_1]
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- k = cost of risk (a dollarized Sharpe ratio). σ = dollar risk measure, not necessarily a standard deviation.
- Measure returns in dollars, not percent. Percent returns mean nothing for margin trades, short sales, futures and short options.
- The textbook CAPM form V0 = μ / (1+r+π) is the special case where the risk charge πV0 is proportional to value. This breaks down at high risk (the Elon Musk coin-flip example). The general form is V0 = (μ − kσ) / (1+r).

GVE solutions to know (risk-free, k = 0)

#	Cash flow pattern	Capital gain E[V1] − V0	Value V0
1	Single payment C1	−V0	C1 / (1+r)
2	Flat perpetuity C	0	C / r
3	Growing perpetuity C, g	gV0	C / (r − g), needs r > g
4	Flat annuity, n payments	−C / (1+r)^n	(C/r)[1 − 1/(1+r)^n]
5	Growing annuity	gV0 − (1+g)C0(1+g)^n/(1+r)^n	(C0/r*)[1 − 1/(1+r*)^n], r* = (r−g)/(1+g)
6	Linear perpetuity {0,1,2,3,…}	1/r	1/r² (links to duration)
7	Quadratic perpetuity {0,1,4,9,…}	1/r + 2/r²	(1/r)(1/r + 2/r²) (links to convexity)
8	Fibonacci {0,1,1,2,3,5,…}	(1+V0)/(1+r)	1 / (r² + r − 1), needs 1+r > 1.618

- Loan amortization is the GVE: End balance = Begin + interest − payment.
- Forward contract via GVE: E[S_T] cancels on both sides, so risk-neutral pricing falls out. F0 = S0 e^(rT) − d_T, where d_T = future value of known dividends.

- Single risky cash flow C1 ~ N(μ, σ): V0 = (μ − kσ) / (1+r). This is certainty-equivalent valuation and the basis for risk-neutral option pricing.
- Duration and convexity: the first and second derivatives of bond value with respect to r give linear-growth and quadratic-growth annuities.

Check yourself

- [ ] Derive the perpetuity value two ways: geometric series and GVE
- [ ] Write the GVE for a growing annuity and solve for V0
- [ ] Explain why a 0-mean, high-risk deal can have negative value

Week 2 – Loans, bonds and the term structure

Risk-free fixed income is valued by discounting each payment at its own zero-coupon rate. Default turns a loan's expected payments into a decaying annuity.

Material: Lecture 2A (Loans with exponential default risk), Lecture 2B (Bonds and rates), VFE Ch 3–4. Mini Project 1 (rent vs buy) and Mini Project 2 (ZCB term structure) used this material.

Loan amortization (Excel)

- Payment = PMT(rate/12, nper, −loan). Example: $5,000, 24 months, 6% → $221.60
- Interest = begin balance × rate/12. Principal reduction = payment − interest. End = begin + interest − payment.
- Balance at any month = PV of remaining payments. Example: month 15 balance = −PV(0.005, 9, 221.60) = $1,945.47
- Total interest = payment × number of payments − loan amount
- Excel functions: PV, PMT, NPER, RATE, FV, NPV (first flow in one period), IRR (first flow immediate)

Loans with constant default rate π (no recovery)

- Expected payments decline at g = −π, so value the loan as a growing annuity
- Example: bank cost 0.1%/month, π = 20 bp/month → loan worth $5,123.73, profit $123.73
- Annual YTM = monthly YTM × 12 (no compounding; the standard for loans and bonds)

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r^* = \frac{r + \pi}{1 - \pi}, \qquad \text{credit spread} = r^* - r \approx \pi
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Bonds, spot and forward rates

- Idealized bond: annual coupon ÷ payments per year (not compounded). Example: Microsoft 6%, 5 years, quarterly → $1.50 a quarter
- Forward rate from no arbitrage: $(1 + r02)^2 = (1 + r0)(1 + f1)$. Cheat-sheet example: $1.022^2 = 1.02(1 + f) \rightarrow f = 2.4004\%$
- Continuous discounting: the spot rate is the average of the forward rates
- Price a forward bond by discounting with forward rates
- Term structure theories: expectations hypothesis (rejected by data), market segmentation (untestable), preferred habitat and liquidity preference (consistent with data). Forward rate = expected future spot + term premium.

Floating rate notes

- Idealized FRN is worth par F on each reset date (GVE: $r0V0 = (F - V0) + r0F$ gives $V0 = F$)
- Interest rate risk of an FRN ≈ a ZCB maturing at the next reset
- A single unknown coupon r1F: replace the unknown rate with its forward rate, then discount at the risk-free rate (slide 13: $V0 = f1\ F\ e^{(-r0)}$)
- Fixed bond: fixed payments, changing value. FRN: changing payments, roughly fixed value.

Stochastic rates (binomial ZCB)

- $r1 = r0 + \mu r \pm \sigma r$ with probability 1/2 each. Solve backward: V1 known in each state, then V0 from the GVE.
- Risk measure = volatility of the bond price, not the rate. k = liquidity risk premium.
- Result: $f1 \approx E[r1] + k\sigma r$. The convexity effect (k = 0) is tiny, of order σr^2.

Treasury instruments

- T-bill: price = face × (1 − discount yield × days/360). Example: 43 days, 5.265% asked → $9,937.11, continuous yield 5.355%
- STRIPS: Treasury coupon and principal sold separately as ZCBs, quoted per $100 face
- Bootstrapping: fit discount factors $d(t) = e^{(-rt)}$ for each payment date. Use a smooth function, regression, or splines. Quotes are clean prices (no accrued interest).
- Basis point = 0.01%. Treasuries pay semiannually, US corporates quarterly, mortgages monthly.

Inflation

- Fisher equation. Discrete: $\rho = (r - i)/(1 + i)$. Continuous: $\rho = r - i$

- Discount nominal cash flows at nominal rates, real cash flows at real rates
- TIPS principal adjusts with CPI and has a deflation floor at par. Implied inflation ≈ Treasury yield − TIPS yield.

Check yourself

- [ ] Build a 24-month amortization table from memory
- [ ] Show why the YTM of a loan with constant default rate equals r*
- [ ] Compute f1 from two spot rates, discrete and continuous
- [ ] Convert a T-bill quote to price and continuous yield

Week 3 – Events, corporate bonds and securitization

An event with constant probability per period shows up in the GVE as an expected loss in the capital gain or cash flow term. A corporate yield = risk-free rate + expected loss + extra risk premium.

Material: Lecture 3 (Events, Corporate Bonds and CDOs). Mini Project 3 (CDOs) used this material.

Event models with the GVE (no risk charge)

Case	GVE	Value
Auto insurance, pays C with prob π each year, perpetual	$rV0 = 0 + \pi C$	$\pi C / r$
Life insurance, pays B once, death prob δ per year	$rV0 = 0 + \delta(B - V0)$	$\delta B / (r + \delta)$
Corporate bond, default prob π, loss fraction λ	expected payments decay at π	$IRR = (r + \pi)/(1 - \pi)$

- Termination risk works like a higher discount rate $(r + \delta)$
- BIS defaultable bond example: face 1,000, 5.5% coupon, $\lambda = 60\%$, $\pi = 3\%$, $r = 4\%$ → value $984.73, YTM 5.86%, spread 1.86% ≈ $\pi\lambda/(1 - \pi)$

Credit spreads

- One period, no risk premium: value = $(1 - \pi\lambda)/(1 + r) \approx 1/(1 + r + \pi\lambda)$. Exact spread = $\pi\lambda(1 + r)/(1 - \pi\lambda)$
- With a market risk premium k: promised yield = $r + k + \pi\lambda$, credit spread = $k + \pi\lambda$

- PD = π, LGD = λ, expected loss EL = πλ
- Citigroup decomposition: YTM 2.42% − risk-free 1.39% = 1.03% spread. EL = 0.22% × 60% = 0.13%. Extra risk premium = 0.90%
- Investment grade = Moody's Baa / S&P BBB and above. Below that is high yield.
- Bonds vs loans: bonds are public with a trustee and indenture. Loans are private, heavily monitored, with covenants and lower LGD.
- Callable bond = straight bond − issuer's call. Convertible = straight bond + investor's call. OAS = yield with options − yield without.

Simulating default times

- Hard way: draw υ each period, default if υ < π
- Easy way (geometric distribution): t = ln(1 − υ) / ln(1 − π)
- Continuous (Poisson / exponential): t = −ln(1 − υ) / λ. Example: λ = 0.4, υ = 0.69 → t = 2.93

Securitization

- Turns risky cash flows into investable assets. A trust holds the assets and pays out cash flows, often in tranches by priority.
- Credit enhancements: subordination, overcollateralization, excess spread. Diversification across the pool also cuts risk.
- Why securitize: specialization, a different risk profile, less equity capital needed, observable prices, and higher total value of the cash flows.
- MBS = mortgage pools. CMOs split MBS cash flows into tranches. The CMO market's failure was the proximate cause of the 2008 crisis.

Interest rate parity (appendix)

- F = S × (1 + r_domestic) / (1 + r_foreign). Example: 1.30 USD/EUR, USD 10%, EUR 8% → F = 1.30 × 1.10 / 1.08 = 1.32 USD/EUR
- The higher-rate currency trades at a forward discount

Check yourself

- [ ] Derive the life insurance value δB/(r + δ) from the GVE
- [ ] Split a corporate yield into risk-free, expected loss and risk premium
- [ ] Simulate a default time from a uniform draw

Week 4 – Simulation

Any random variable can be simulated as $F^{-1}(u)$ with u uniform on [0, 1]. Always report a simulated estimate as the mean ± 2 standard errors.

Material: Lecture 4 (Simulation), Valuation Principles Ch 5–6, Cholesky five and Simulation spreadsheets (Excel).

Inverse CDF method

- Exponential: $t = -\ln(1 - u)/\lambda$
- Normal: z = NORM.S.INV(RAND())
- Simulation replaces integration: $E[g(u)] = \int_0^1 g(u)\,du \approx (1/n)\,\Sigma\, g(ui)$. Essential in many dimensions.

Every simulation, step by step (memorize)

1. Draw N random numbers or vectors z
2. Compute outcomes f(z)
3. Mean $\bar{X}$ = estimate of E[f(z)]
4. Standard deviation s, standard error $s/\sqrt{N}$
5. Report $\bar{X} \pm 2s/\sqrt{N}$ (the class uses m = 2)
6. To halve the error, use 4× the sims. Error shrinks with $\sqrt{N}$.

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E[f(z)] = \bar{X} \pm 2\frac{s}{\sqrt{N}}
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- Oil example: N ~ Normal(1500, 500) barrels, P ~ Exponential(0.02). Revenue: 70,779 ± 4,810 (1,000 sims) vs 75,036 ± 160 (1,000,000 sims)

Variance reduction (example: E[e^z] = e^0.5 = 1.6487)

Technique	What you do	Effect
More sims	Increase N	Max error falls with $\sqrt{N}$
Moment matching	Rescale z to mean 0, sd 1	Better estimate (1.66 vs 1.59), error not necessarily smaller
Antithetic sampling	Add −z for every z	Lower error, forces mean and skew to 0

Technique	What you do	Effect
Control variate	Regress e^z on z and z^2, use known E[z] = 0, $E[z^2] = 1$	Biggest gain: max error 0.014 at 1,000 sims vs 0.13–0.15 for the others

- Control variate estimate = b0 + b1·0 + b2·1 (example: 0.84 + 0.80 = 1.64)
- Variance of the mean response = $s_e^2\ X_h'(X'X)^{-1}X_h$, with $X_h = [1, 0, 1]$. Coefficient covariance = $s_e^2(X'X)^{-1}$
- OLS: $b = (X'X)^{-1}X'y$ (the normal equations). Know LINEST output rows.

Correlated variables

- Two normals with correlation ρ: $z2 = \rho z1 + z3\sqrt{(1-\rho^2)}$, z3 independent of z1
- Many normals: multiply independent draws by the Cholesky factor of the target correlation/covariance matrix
- Pearson correlation describes linear (normal) dependence only
- Copula: make correlated normals, map to uniforms with N(z), then apply each $F^{-1}(u)$. Lets you "correlate" any distributions, e.g. two exponentials with λ = 0.2 and 0.3
- Matrix review: $1'\Sigma 1$ = variance of the equal-weight sum, $a'\Sigma a$ = portfolio variance, $a'\Sigma b$ = covariance of two portfolios

Check yourself

- [ ] Write the 6-step simulation recipe and the ±2 SE rule
- [ ] Explain why a control variate beats antithetic sampling
- [ ] Generate two correlated exponentials with a copula

Week 5 – Stochastic processes

Four building blocks: arithmetic (ABM) for spreads and rates, geometric (GBM) for asset prices, mean reversion (MR) for rates, commodities and FX, and jumps added on top of any of them. These moments are on the cheat sheet; know which process fits which variable.

Material: Lecture 5 (Stochastic Processes), Week 5 simulation data and examples (Excel).

Process	One-step update	E[S_t+j given S_t]	Var[S_t+j given S_t]	Used for
ABM	$S_{t+1} = S_t + \mu + \sigma z$	$S_t + j\mu$	$j\sigma^2$	Spreads, rates, can go negative
GBM	$S_{t+j} = S_t e^{((\alpha - \sigma^2/2)j + z\sigma\sqrt{j})}$	$S_t e^{(\alpha j)}$	ln S: $\sigma^2 j$	Asset prices, never hits zero
MR (OU)	$S_{t+1} = S_t + \kappa(\mu - S_t) + \sigma z$	$\mu + (S_t - \mu)(1 - \kappa)^j$	$\sigma^2[1 - (1 - \kappa)^{(2j)}] / [\kappa(2 - \kappa)]$	Rates, commodities, FX

- Covariances: ABM $\text{Cov}(S_{t+j}, S_{t+m}) = \sigma^2 \min(j, m)$. GBM same for ln S. MR: $\text{Var}(S_{t+j}) \times (1 - \kappa)^{(m-j)}$ for $m \geq j$
- GBM log mean: $E[\ln S_{t+j}] = \ln S_t + (\alpha - \sigma^2/2)\, j$
- MR with $\kappa = 0$ becomes ABM. With $\kappa = 1$ it jumps straight to the mean each step.
- Jumps: each period, with probability π, add J. J and π can be random or depend on the state.
- Links to Week 1: ABM ~ linear growth perpetuity, GBM ~ exponential growth perpetuity, MR ~ mean-reverting perpetuity

Parameter estimation

- GBM: take log returns, compute their mean and sd. Drift α = mean log return + $\sigma^2/2$. Scale mean by T and volatility by $\sqrt{T}$.
- MR: regress ΔX on X. Constant = $\kappa\mu$, slope = $-\kappa$. So $\kappa = -\text{slope}$, μ = constant/κ. Check the t-stat on the slope to judge whether mean reversion is real (class example: MORTGAGE30US).

Continuous-time versions

$$dS = \alpha S\,dt + \sigma S\,dz \quad (\text{GBM}), \qquad dX = \kappa(\mu - X)\,dt + \sigma\,dz \quad (\text{OU})$$

- Wiener process: $E[dz] = 0$, $\text{Var}[dz] = dt$
- Itô's lemma for $dW = \alpha\, dt + \sigma\, dz$: $df = (\alpha f_W + \tfrac{1}{2}\sigma^2 f_{WW} + f_t)\, dt + \sigma f_W\, dz$

Optimization with simulation

- Tree-harvest example: price follows ABM, harvest once when price first hits a threshold before T. Find the threshold that maximizes the simulated PV.

- Use fixed random numbers (fixed seeds) across the candidate rules so the comparison is fair.

Check yourself

- [ ] Pick the right process for a stock, a credit spread and an interest rate
- [ ] Compute E and Var of an MR process 12 steps ahead
- [ ] Estimate κ and μ from a regression output

Week 6 – MPT and the CAPM

MPT finds the minimum-variance and tangency portfolios from given returns and covariances. Sharpe's CAPM prices risk by beta. Lintner's version derives prices from cash flows and is the GVE in disguise.

Material: Lecture 6 (MPT and the CAPM). This is this week's lecture, so expect a share of the exam; the recitation questions on slides 54–58 are good practice.

Leverage and the Sharpe ratio

- Sharpe ratio = $(r_i - r)/\sigma_i$. Dollarized Sharpe = (expected $ return − risk-free funding cost)/$ sd. Use the dollar version for shorts and futures.
- M&M Prop 2: $r_E = r + (1 + D/E)(r_U - r)$, $\sigma_E = (1 + D/E)\sigma_U$. Risk-free leverage leaves the Sharpe ratio unchanged.
- Example: $r_U = 10\%$, $r = 4\%$, $\sigma_U = 15\%$ → Sharpe 0.40. With 25% equity ($D/E = 3$): $r_E = 28\%$, $\sigma_E = 60\%$, Sharpe still 0.40

Portfolio math

- Two assets: $r_p = w_A r_A + w_B r_B$. $\sigma_p^2 = w_A^2\sigma_A^2 + w_B^2\sigma_B^2 + 2\rho\, w_A w_B \sigma_A \sigma_B$
- Lower correlation = more diversification. $\rho = -1$ allows a perfect hedge.
- Risky asset + risk-free asset = straight line. The best line touches the frontier at the tangency portfolio (maximum Sharpe ratio).
- n assets: mean = $r'w$, sd = $\sqrt{(w'\Sigma w)}$, weights sum to 1, shorts allowed

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A = \mathbf{1}'\Sigma^{-1}\mathbf{1},\; B = \mathbf{1}'\Sigma^{-1}r,\; C = r'\Sigma^{-1}r,\; D = AC - B^2
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\sigma^2(R) = \frac{AR^2 - 2BR + C}{D}, \qquad r_T = \frac{C - Br_f}{B - Ar_f}, \qquad w_T = \frac{\Sigma^{-1}(r - r_f\mathbf{1})}{B - Ar_f}
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- Excel: maximize Sharpe with Solver, variance = MMULT(TRANSPOSE(w), MMULT(cov, w)), betas = MMULT(cov, w)/portfolio variance
- MPT on historical data gives extreme, unstable weights because sample means are noisy

Sharpe CAPM

- $r_i = r_f + \beta_i(r_m - r_f)$, $\beta_i = \text{cov}(r_i, r_m)/\sigma_m^2$ (the security market line)
- Total variance: $\sigma_i^2 = \beta_i^2\sigma_m^2 + \sigma_\varepsilon^2$. Idiosyncratic risk has no price in Sharpe's CAPM.
- SML = return vs beta. CML = return vs sigma (the tangency line). Positive alpha = above the SML.
- Beta in practice: SLOPE(stock returns, index returns). Alpha = INTERCEPT − $r_f(1 - \beta)$
- Valuation example: $\beta = 0.8$, $r_f = 3\%$, market premium 6% → 7.8%. Dividend \$0.50 growing 2% → 0.50/(0.078 − 0.02) = \$8.62
- Extensions: Black zero-beta, Brennan tax, Merton ICAPM, Breeden consumption CAPM, international CAPM, Fama-French 3-factor, Carhart 4-factor
- APT: arbitrage-based, $r_i = r_f + \beta_{i1}\lambda_1 + \beta_{i2}\lambda_2 + \ldots$ Factors are specified up front or found by factor analysis / PCA (hard to interpret). BARRA is a commercial version.

Lintner CAPM (cash-flow version)

- Investor maximizes $N'\mu + (W0 - N'V)(1 + r) - \frac{1}{2}A\,N'\Sigma N$. Setting $N = 1$ (markets clear) gives prices:

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V = \frac{\mu - A\,\Sigma\mathbf{1}}{1 + r}, \qquad A = \frac{\mathbf{1}'(\mu - V(1+r))}{\mathbf{1}'\Sigma\mathbf{1}}
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- Single asset: $V = (\mu - A\sigma^2)/(1 + r)$. This is the GVE with variance as the risk measure, and it allows negative values for very risky assets.
- Derivatives (zero net supply) are priced at $V = B\mu$: risk-neutral.
- Risk charge for each asset = A × its covariance with the market. Equivalent CAPM form: $V0 = [\mu - \lambda\,\text{cov}(C1, r_m)]/(1 + r)$, $\lambda = (r_m - r)/\sigma_m^2$
- Class example: means 10, 20, 30, A = 0.06, r = 0 → expected returns 31.58%, 26.10%, 18.76%, market 23.15%, betas 1.36, 1.13, 0.81
- Key differences from Sharpe: idiosyncratic variance does affect prices, and lower correlation raises asset values (so expected % returns fall)
- CDOs can be valued this way: allocate the collateral's total risk charge to tranches by cov/var

Check yourself

- [ ] Compute a levered Sharpe ratio and show it is unchanged
- [ ] Find A, B, C, D and the tangency portfolio for a 3-asset case
- [ ] Value a growing dividend using a CAPM discount rate
- [ ] Explain how Lintner's CAPM prices idiosyncratic risk